

MEDLAB

MIS102 Data & Networking
Action Plan

Assessment 3

AI Declaration:

We declare that except where we have referenced, the work we are submitting for this assessment task is our own work. We have read and are aware of Torrens University Australia Academic Integrity Policy and Procedure, viewable online at http://www.torrens.edu.au/policies-and-forms. We are aware that we need to keep a copy of all submitted material and their drafts, and we will do so accordingly.

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Introduction

MedLab is a new medical laboratory established in New Zealand that requires assistance in implementing their network. As a consultant for MedLab, in this action plan I will discuss the creation of the network, including physical media, high-level network topology, internet access, storage, and recommendations to ensure an effective network. I will also discuss network administration and performance, which will be supported by a tool that help administer the network. A network disaster recovery plan is also developed for two scenarios: internet outage and malware attack.

MedLab is assumed to manage large data such as x-rays, (CT/MRI) scans, and genetic data, which provide vital information for healthcare providers. It is assumed since MedLab is newly established, they will require scaling capabilities, demanding more devices on the network.

Network Setup

I have decided to recommend an extended star topology (*Figure 1*). This segments the network through switches one segment for computers and servers, adding redundancy, easier network maintenance along with integration of new devices to the network. The devices on the network will be connected through shielded gigabit ethernet cables (cat 6a/7a), allowing up to 10GB/s of data throughput also protecting from electromagnetic interference (medical scanner machines). This type of cabling is necessary to keep data intact, allow large file transfers, future-proof, and keep the system dependable.

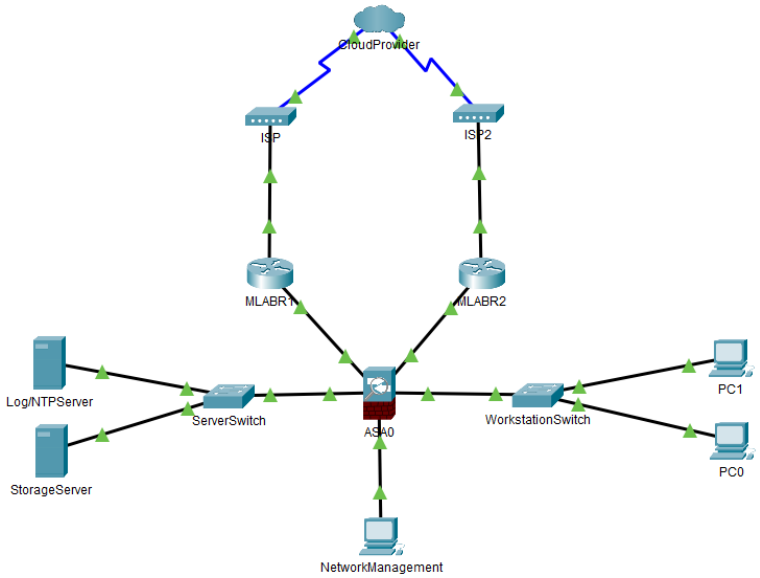


Figure 1: MedLab proposed network structure.

It is important to have as much uptime, security, and speed as possible. Taking these things into account, Wi-Fi capable modems connected through shielded gigabit ethernet cables (cat 6a/7a) utilizing multi-homing is my recommended internet connection. Although Rafaqat & Muhammad (Khan & Tariq, 2019) boast that wireless is the better medium, I would recommend that the wireless capability the modems are disabled until necessary for business operations. Having a strictly wired system increases security, lowers network attack vectors, provides reliability, and increased data rates. Although disabling wireless capabilities restricts transfer mediums, mobility, and limits flexibility, it can be activated and carefully integrated if needed in the future. Establishing a secure and reliable network is important for a newly established business, meanwhile having the option to enable wireless capabilities enables MedLab to enhance their operations in the future.

Multi-homing is a method to connect the local area network (LAN) to multiple ISP's, see *Figure 1*. The multi-homing method removes the single point of failure issue for internet, providing redundancy to the network connectivity.

In terms of storing data, an on-premises storage server accompanied by a cloud storage backup server are recommended.

The on-premises server will be a base for storing all necessary business operation data, which will then be backed up to the cloud. The on-premises server must be secured using encryption methods, data authorization controls, access control lists, and hashing methods among other to ensure compliance with data storage regulations (Privacy Commissioner, 2020) in New Zealand.

Supplying a cloud server will provide MedLab with external backups, allowing for re-establishment of critical data in event of corruption or a disaster. The effectiveness of cloud backups in disaster recovery along with considerations are elaborated on in this report by Muhammad and others (Hasan et al., 2023).

The log/NTP server will assist in logging and time normalization between network devices, where the proposed network monitoring tool (discussed in the Network Administration & Performance section) can be installed and maintained from utilizing the Network Management computer (see *Figure 1*).

Network Administration & Performance

Logging tools can centralise several types of logs (system, network, security, browser, etc) from network devices, giving a centralized hub for the network administrators to analyse network operations. Performance tools also enhance networks by allowing monitoring of network traffic, device health and performance, bandwidth, uptime, and prevention methods.

In a medical laboratory, logging and monitoring tools are essential for protecting network integrity and business operation continuity. Logging and monitoring tools do this through assistance with issue detection and supplying preventative maintenance suggestions. Without performance and logging tools, networks can fail through data-breaches, downtime, and reduce productivity, causing severe harm to MedLab's reputation and financial costs.

In consideration of the need for a network administration tool, am recommending PRTG network monitor. The PRTG tool provides a free, open-source monitoring & logging solution for network management. Key features of PRTG are automatic device discovery, alerts, reports, alarms, maps (network visualization), and serverless. Along with monitoring of hardware, software, network performance, and data traffic.

PRTG allows a 30-day free trail of the full version, with options to upgrade to the fully paid version with more features or utilize the free version with limited features. The free tier provides small businesses, like MedLab, with a cost-effective solution with the option for scaling.

For example, the Austrian Red Cross uses PRTG to monitor critical systems involved in blood donation, collection, management, and distribution, emphasizing it's reliability in mission-critical environments (Paessler, 2023). Another success story is Dayton Children's Hospital (DCH) in the U.S, with the addition of PRTG as a network monitoring software DCH improved their productivity, reducing the need to dump resources into managing the IT infrastructure, increasing uptime to 99.99%, averaging 2.4 minutes of unscheduled downtime (Paessler, 2024).

Network Disaster Recovery Plan (NDRP) Advice

It is important to remember that a NDRP is part of multiple business documents that need all culminate into robust disaster recovery plan. Some extra documents needing to be considered consist of these plans:

- business continuity (BCP)
- risk management (RMP)
- backup plan
- business impact analysis (BIA)

These additional documents should be explored by management in conjunction with the advised NDRP in the future. It is particularly important to provide regular updates, training, and testing, as explained in this article on NDRP's (Schultz, 2024).

Updating the documentation will address emerging threats and specific challenges related to MedLab. Training will allow employees to navigate the documentation by ensuring they know their allocated roles in a disaster situation. Testing is extremely important, as it is one thing to create documentation, but critical to ensure that employees can work together and know their roles in a disastrous event.

If these supporting documents are created, updated, and maintained, along with providing employees with training and testing situations, MedLab will be equipped to recover from disasters in a timely and cost-effective manner.

Difana and collaborators (Meilani et al., 2019) create a full-fledged disaster recovery plan for a university, outlining the necessity for additional supporting documents to create a robust NDRP with metrics like recovery point objectives (RPO) and recovery time objectives (RTO) within a business continuity plan.

In event of either of the two proposed disasters, The NDRP will supply employees with actionable steps to assist in recovering from each event. This should not represent the final NDRP. See ***Appendix A*** for the relevant NDRP table.

Conclusion

In conclusion, I have given advice on the network topology by suggesting an extended star topology that enhances redundancy. I recommended to utilize shielded ethernet cables in consideration of the laboratory setting, minimizing electromagnetic interference with connections. I suggested to utilize multi-homing methods to mitigate the single point of failure issue for internet connection and suggested a cloud server to enhance data continuity and provide reliable backup solutions.

The PRTG network monitoring tool will provide a comprehensive and centralised network monitoring tool, enhancing administrative operations, increasing productivity, preventing issues, and monitoring device and network performance.

I have created a NDRP with actionable steps to recover from an internet outage and malware attack, where regular revision and updates, tests, and training are essential to address emerging threats and ensure employees are prepared for disaster recovery. I also suggested to explore and create supporting documents to create robust documentation for business continuity. If these recommendations are followed and documents are regularly maintained and updated, MedLab will produce a stable network environment that ensures data security, reliability, and accessibility.

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Appendix A. NRDP for Internet outage and Malware attack

	INTERNET OUTAGE	**MALWARE ATTACK**
Event occurred	• Internet connectivity is lost	• Malware is detected on one or more devices
Event Detection/Notification	• Notify IT team of lost internet connectivity	• IT team is notified by employee or suspicious network behaviour
Initial Assessment	• Determine the spread of the internet outage. ○ What devices are affected?	• Identify infected devices. • Isolate infected devices from the network (disconnect from the network, e.g. unplug network cables)
Cause Analysis	• Determine outage causation, scheduled or unscheduled? ○ Contact ISP ○ Use ISP's website (outage notifier)	• Identify malware source. ○ Email, website, file download. • Identify vulnerable entry points. ○ Open ports, malicious websites, malicious files • Hire independent cybersecurity team to analyse cause
Recovery	• Use multi-homing method to utilize second ISP. • Scheduled outage ○ Follow ISP instructions to recover connection. • Unscheduled outage ○ Restart routers	• Use anti-virus/malware to remove source file. • Block malicious website on network. • Restore data from backups (cloud backups)
Testing	• Ping from each device to the internet • Ensure network devices can communicate with each other. • Verify critical applications and services are operational	• Run malware scans on affected devices. • Run network-wide scan. • Ensure NO malware is present when re-connecting to network
Documentation	• Document the outage. ○ Time, date, downtime • Explore additional solutions to learn from event and note them.	• Document the attack. ○ Time, date, downtime ○ How? ○ What devices were affected? ○ What data was affected? • Explore additional solutions to learn from event and note them.
Communication	• Meetings to re-evaluate business plans. • Presentations to summarize on event. • Notify stakeholders about outage	• Educate employees about attack vectors. ○ Phishing emails ○ Harmful downloads ○ Malicious websites • Notify stakeholders about attack